

Scale Invariant Point Processes and Dickman Convergence

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Sum of Records

For a random permutation $\sigma \in \Pi_n$, let S_n be the sum of positions of records which are positions k such that $\sigma(k) > \sigma(1), \dots, \sigma(k-1)$.

One can show that $S_n =_d \sum_{k=1}^n k B_k$ with $B_k \sim \text{Ber}(1/k)$ independent.

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For $(U_i)_{i \geq 1}$ a sequence of i.i.d. $\mathbb{U}[0, 1]$ random variables, the distribution of the random variable

$$D = U_1 + U_1U_2 + \cdots = \sum_{k=1}^{\infty} \prod_{i=1}^k U_i$$

is called the *standard Dickman distribution*.

For D a standard Dickman random variable, $S_n/n \xrightarrow{d} D$.

Dickman Distribution

First appeared in Dickman'30 in the context of *smooth numbers*.

For x and y positive, let

$$\psi(x, y) = \# \text{ of integers } \leq x \text{ with all its prime factors } \leq y.$$

Then for all $u > 0$,

$$\psi(x, x^{1/u}) \sim x\rho(u) \text{ as } x \rightarrow \infty$$

where ρ is the Dickman density function up to a scaling.

Since then, has appeared in logarithmic combinatorial structures, algorithms, probabilistic number theory etc.

Dickman Convergences

- Let $X = \{X(t) : t \geq 0\}$ be a Gamma process with Lévy measure $\nu(x) = x^{-1}e^{-x}, x > 0$ and X_ϵ be the process with jumps of X not exceeding $\epsilon > 0$.

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- For $p_n, n \geq 1$ an enumeration of primes, let Ω_n be the set of integers with largest possible prime factor p_n .

Consider a random variable $M_n \in \Omega_n$ with $\mathbf{P}\{M_n = x\} \propto 1/x$.

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Can one characterize these convergences?

A Second Characterization

Let η be a scale invariant Poisson point process on $\mathbb{R}_+$ with intensity measure $x^{-1}dx$. Then one can show

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Heuristically, if a sequence of point processes $(\xi_n)_{n \geq 1}$ on $\mathbb{R}_+$ converges to η , then under some suitable conditions, one can expect that

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Example: Let $\xi_n = \sum_{k=1}^{\infty} B_k \delta_{k/n}$ with $B_k \sim \text{Ber}(1/k)$ independent.

One can check that $\xi_n \xrightarrow{vd} \eta$ and

$$n^{-1} \sum_{k=1}^n k B_k = \int_0^1 t d\xi_n \xrightarrow{d} D.$$

Convergence to Poisson Process

Let $(\xi_n)_{n \geq 1}$ be a sequence of point processes. Then $\xi_n \xrightarrow{vd} \eta$ iff for all $0 < a < b < \infty$,

- ① $\lim_{n \rightarrow \infty} \mathbf{P}\{\xi_n[a, b] = 0\} = \frac{a}{b},$
- ② $\liminf_{n \rightarrow \infty} \mathbf{P}\{\xi_n[a, b] \leq 1\} \geq \frac{a}{b} \left(1 + \log \frac{b}{a}\right).$

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Sum of small points (B., Molchanov)

Let $\xi_n \xrightarrow{vd} \eta$ and satisfies in addition that

③ $\lim_{\epsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \mathbb{E} \int_0^\epsilon t d\xi_n = 0.$

Then,

$$\int_0^1 t d\xi_n \xrightarrow{d} \int_0^1 t d\eta =_d D.$$

Small Jumps in a Gamma Process

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Using that jumps occur as a Poisson process, letting $\Pi_{X(t)}$ denote the process of jump sizes, it follows $\xi_n = n\Pi_{X(1)}$ satisfies conditions (1)-(2) and hence as $\epsilon \rightarrow 0$,

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Moreover, one also has that

$$\limsup_{n \rightarrow \infty} \mathbb{E} \int_0^\epsilon t d\xi_n = \epsilon$$

and hence we recover Covo's result that as $n \rightarrow \infty$,

$$\epsilon^{-1}X_\epsilon(1) =_d \int_0^1 t d\xi_n \xrightarrow{d} D$$

as a consequence of the convergence of the underlying point processes.

Reciprocal Distribution on Ω_n

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For $X_k \sim \text{Geom}(1 - 1/p_k)$, one can show $M_n =_d \prod_{k=1}^n p_k^{X_k}$ and thus

$$\frac{\log M_n}{\log(p_n)} =_d \frac{1}{\log(p_n)} \sum_{k=1}^n X_k \log(p_k) = \int_0^1 t d\xi_n$$

where

$$\xi_n =_d \sum_{k=1}^{\infty} X_k \delta_{\log p_k / \log p_n}.$$

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One can check that $(\xi_n)_{n \geq 1}$ satisfies conditions (1)-(3). Hence

$$\xi_n \xrightarrow{vd} \eta \quad \text{and} \quad (\log p_n)^{-1} \log M_n \xrightarrow{d} D.$$

Additional Randomness: Uplifting

Let $(\xi_n)_{n \geq 1}$ be a sequence of simple point processes with $\xi_n \xrightarrow{vd} \eta$. Denote by $(x_k^n)_{k \geq 1}$ an enumeration of the points in ξ_n .

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For $U, U_i, i \geq 1$ i.i.d. random vectors in $\mathbb{R}^d$, consider the sequence of random measures $(\xi_n^U)_{n \geq 1}$ obtained by uplifting the points of ξ_n to $\mathbb{R}^d$ i.e.

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Since $\xi_n \xrightarrow{vd} \eta$, one can expect that

$$\xi_n^U \xrightarrow{vd} \eta^U =_d \sum_{k=1}^{\infty} \delta_{U_k x_k}$$

where $(x_k)_{k \geq 1}$ is an enumeration of η . Note here that η^U is also scale invariant.

Compact vs. non-compact support

If U has a relatively compact support, then one indeed has $\xi_n^U \xrightarrow{vd} \eta^U$.

Follows by noting that for f a bounded continuous function with relatively compact support, the function

$$h(x) = -\log \mathbb{E}e^{-f(Ux)}$$

also has relatively compact support and thus

$$\mathbb{E}e^{-\xi_n h} \rightarrow \mathbb{E}e^{-\eta h}. \quad (\star)$$

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For U possibly unbounded, h may not anymore be compactly supported. Thus, one needs to generalize $(\star)$ for such functions.

Convergence for non-compact

Let $(\xi_n)_{n \geq 1}$ and ξ be point processes such that $\xi_n \xrightarrow{vd} \xi$ for a simple ξ satisfying $\mathbb{E}\xi(\partial B_r) = 0$ for all $r > 0$.

We require the function h and $(\xi_n)_{n \geq 1}$ to satisfy the following:

$$\begin{cases} \lim_{r \rightarrow 0} \limsup_{n \rightarrow \infty} \mathbb{E} \int_{B_r} h(x) d\xi_n = 0, \\ \lim_{r \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{E} \int_{B_r^c} h(x) d\xi_n = 0, \end{cases}$$

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Convergence in case of non-compact support (B., Molchanov)

For a non-negative bounded continuous function h satisfying the above conditions,

$$\mathbb{E}e^{-\xi_n h} \rightarrow \mathbb{E}e^{-\xi h}.$$

Further Directions

The conditions imposed on the processes can possibly be relaxed a bit.

There are other examples, e.g. in the context of *minimal directed spanning trees* where one has Dickman convergence. It would be nice to unify all such examples under this general set up.

It would be interesting to study the higher dimensional Dickman type random variables obtained by summing the points of a scale invariant process near zero.

Thank you